Development and structure of Prometheus: the Canadian **wildland fire** growth **simulation model**

C Tymstra, RW Bryce, BM Wotton, SW Taylor… - … Report NOR-X-417 …, 2010 - spyd.com

… • to **review** and evaluate existing models; • to … a **wildfire** at many points around the fire perimeter, including the underlying rates of **spread** for the spatial **simulation** of firefront **propagation**. …

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Modeling **wildfire spread** with an irregular graph network

W Jiang, F Wang, G Su, X Li, G Wang, X Zheng… - Fire, 2022 - mdpi.com

… In this paper, we propose a **wildfire spread model** with an irregular graph … **wildfire spread model** with an irregular graph network (IGN) to simulate and analyze the spatiotemporal **spread** …

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Climate change impact on future **wildfire** danger and activity in southern Europe: a **review**

J Dupuy, H Fargeon, N Martin-StPaul, F Pimont… - Annals of Forest …, 2020 - Springer

… indices describe the potential for fire **spread** (the Initial **Spread** Index, ISI), … **spread** in SPITFIRE, while the maximum daily Nesterov index is used to calibrate the annual burnt area **model** …

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A conceptual high level multi-agent system for **wildfire management**

RB Zadeh, A Elmi, V Moghaddam… - … on Geoscience and …, 2025 - ieeexplore.ieee.org

… This literature **review** investigates previous works, each offering unique perspectives and … early detection and effective monitoring of **wildfire spread**, considering energy consumption of …

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A **review** of traffic models for wildland-urban interface **wildfire** evacuation

A Bergstedt - LUTVDG/TVBB, 2018 - lup.lub.lu.se

… **model** and **review** 12 existing models, both traffic models and evacuation models, and their potential usefulness in WUI **wildfire** … concerning the **spread** and **spread** rate of a **wildfire** is the …

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Distributed **wildfire** surveillance with autonomous aircraft using deep reinforcement learning

KD Julian, MJ Kochenderfer - … of Guidance, Control, and **Dynamics**, 2019 - arc.aiaa.org

… work together to monitor the **wildfire** growth. Furthermore, the … for both **wildfire propagation** and **trajectory** optimization is … **simulation**, the **wildfire** is small, and so no precise **trajectory** …

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Adaptive **forest fire spread simulation** algorithm based on cellular automata

L Sun, C Xu, Y He, Y Zhao, Y Xu, X Rui, H Xu - Forests, 2021 - mdpi.com

… automata with a fixed time step to simulate **forest fire spread** may be limited in its ability to … with an existing **forest fire model** to construct an improved **forest fire spread model**, which …

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A **review** of radiant heat flux models used in **bushfire** applications

AL Sullivan, PF Ellis, IK Knight - … Journal of **Wildland Fire**, 2003 - CSIRO Publishing

… The **prediction** of the thermal radiation from a **wildfire** is … is important in a number of fields, including rate of **spread** modelling, … of heat transfer and fire **propagation**, the physics is reason…

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Federal **forest-fire** policy in the United States

SL Stephens, LW Ruth - Ecological applications, 2005 - Wiley Online Library

… **wildfire** has increased over the last decade. Given the current and future challenges of fire **management**, and based on analytical research and **review** … not contribute to **wildfire spread** or …

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Three lines of defense for **wildfire** risk **management** in electric power grids: A **review**

A Arab, A Khodaei, R Eskandarpour… - IEEE …, 2021 - ieeexplore.ieee.org

… This matter can be formulated through the convective heat transfer terms considered in the **wildfire spread model**. Second, higher winds can also cause the fire front to get closer to the …

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